



LONGEVITY & ANTI-AGING

SS-31

SS-31 – 10 mg per vial or 50 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

|                     |                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| CHEMICAL NAME       | Elamipretide; D-Arg-2',6'-dimethyltyrosyl-Lys-Phe-NH2                                                                                 |
| SYNONYMS / ALIASES  | SS-31; Bendavia; MTP-131; elamipretide                                                                                                |
| CAS NUMBER          | 736992-21-5                                                                                                                           |
| MOLECULAR FORMULA   | C32H49N9O5                                                                                                                            |
| MOLECULAR WEIGHT    | 639.79 g/mol                                                                                                                          |
| AMINO ACID SEQUENCE | D-Arg-Dmt-Lys-Phe-NH2 (4-residue Szeto-Schiller tetrapeptide; Dmt = 2',6'-dimethyl-L-tyrosine; D-Arg at position 1; C-terminal amide) |
| SOURCE              | Synthetic (solid-phase peptide synthesis with non-natural amino acid incorporation)                                                   |

PHYSICAL PROPERTIES

|                      |                                                                              |
|----------------------|------------------------------------------------------------------------------|
| APPEARANCE           | Lyophilized white to off-white powder                                        |
| PURITY SPECIFICATION | >=99% by HPLC (confirm against batch COA)                                    |
| CONTAINER            | 10 mg or 50 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp |

STORAGE & HANDLING

|                                      |                                                                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| UNRECONSTITUTED STORAGE              | -20 C, protect from light and moisture                                                                                          |
| RECONSTITUTED STORAGE                | 2-8 C, stable approximately 28 days in bacteriostatic water                                                                     |
| SUGGESTED RECONSTITUTION SOLVENT     | Bacteriostatic water or sterile water (research use only)                                                                       |
| SHELF LIFE (UNRECONSTITUTED, SEALED) | 24 months at -20 C                                                                                                              |
| STABILITY NOTES                      | The D-amino acid and dimethyltyrosine substitutions confer resistance to enzymatic cleavage. Avoid repeated freeze-thaw cycles. |

## STRUCTURAL NOTES

SS-31 (elamipretide) is a synthetic tetrapeptide developed within the Szeto-Schiller (SS) peptide family. Its sequence is D-Arg-Dmt-Lys-Phe-NH<sub>2</sub>, where Dmt denotes 2',6'-dimethyl-L-tyrosine, a non-natural tyrosine derivative. Two non-natural features distinguish it: a D-arginine at position 1 (rather than the natural L-isomer), and the dimethyltyrosine at position 2. The peptide is amphipathic and cationic, with the dimethyltyrosine contributing aromatic and antioxidant character. SS-31 is classified in the research literature as a mitochondria-targeted peptide that associates with cardiolipin in the inner mitochondrial membrane.

## SELECTED PEER-REVIEWED REFERENCES

1. Szeto H.H. - *First-in-class cardiolipin-protective compound as a therapeutic agent to restore mitochondrial bioenergetics* - British Journal of Pharmacology, 2014 - DOI: 10.1111/bph.12461
2. Birk A.V. et al. - *The mitochondrial-targeted compound SS-31 re-energizes ischemic mitochondria by interacting with cardiolipin* - Journal of the American Society of Nephrology, 2013 - DOI: 10.1681/ASN.2012121216
3. Siegel M.P. et al. - *Mitochondrial-targeted peptide rapidly improves mitochondrial energetics and skeletal muscle performance in aged mice* - Aging Cell, 2013 - DOI: 10.1111/acer.12102

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

## COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.